

Temporal & Multivariate Data

COMP7507

Visualization & Visual Analytics



SCHOOL OF
**COMPUTING &
DATA SCIENCE**
The University of Hong Kong

Types of Data

- **One-dimensional** — linear data
 - e.g., age distribution
 - includes sequential data such as text, program source codes
- **Two-dimensional**: planar or map data
 - e.g., geographical maps, floorplans, newspaper layouts
- **Three-dimensional**: real-world objects
 - e.g. medical scans, architectural design
- **Temporal**
 - e.g. timelines, weather

Types of Data

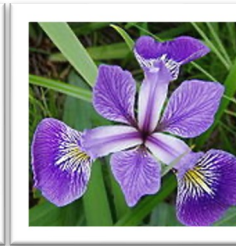
- **Multidimensional** or **Multivariate**
 - e.g., financial data, customer behaviour
- **Tree** (hierarchical data)
 - e.g., file structure, evolution
- **Network** (relational data)
 - e.g., social network, air traffic
- The above classification is not mutually exclusive
 - E.g., how about air traffic data?
 - multivariate, geographical, network, and temporal

The Iris Sample Data Set

- Created by R.A. Fisher
- Possibly the best known data set in the pattern recognition community
- 3 classes (types of iris)
- 50 objects in each class
- 5 attributes
 - sepal length & width (cm)
 - petal length & width (cm)
 - class (Iris Setosa, Iris Versicolour, Iris Virginica)



Iris Setosa



Iris Versicolour



Iris Virginica

[wikipedia]

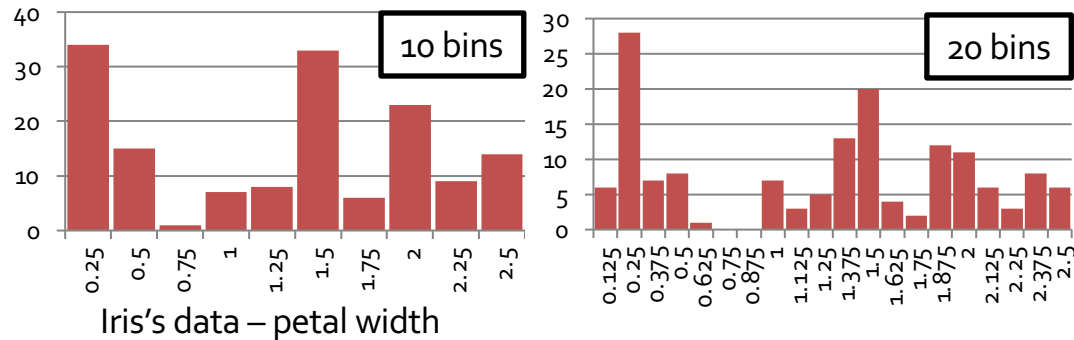
The Iris Sample Data Set

# iris.data Sepal Length	# iris.data Sepal Width	# iris.data Petal Length	# iris.data Petal Width	Abc iris.data Class
5.10000	3.50000	1.40000	0.20000	Iris-setosa
4.90000	3.00000	1.40000	0.20000	Iris-setosa
4.70000	3.20000	1.30000	0.20000	Iris-setosa
4.60000	3.10000	1.50000	0.20000	Iris-setosa
5.00000	3.60000	1.40000	0.20000	Iris-setosa
5.40000	3.90000	1.70000	0.40000	Iris-setosa
4.60000	3.40000	1.40000	0.30000	Iris-setosa
5.00000	3.40000	1.50000	0.20000	Iris-setosa
4.40000	2.90000	1.40000	0.20000	Iris-setosa
4.90000	3.10000	1.50000	0.10000	Iris-setosa

Some Basic Plots

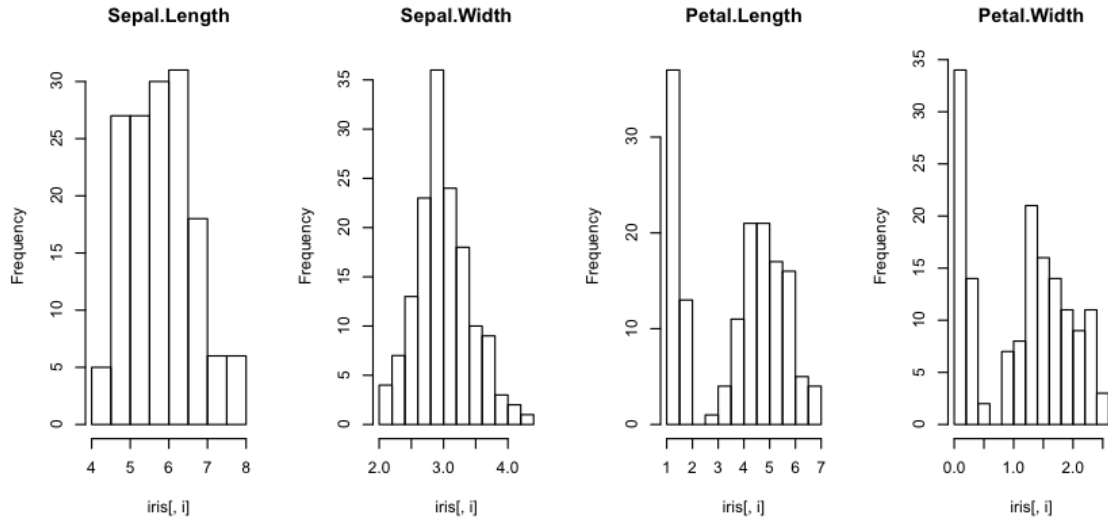
Bar Charts / Histograms

- To show distribution of values of a **single variable**
- Values are divided into **bins**
- A **bar plot** is used so that the height of each bar indicates the number of objects in each bin
- Shape of histogram depends on the number of bins



Box Plots

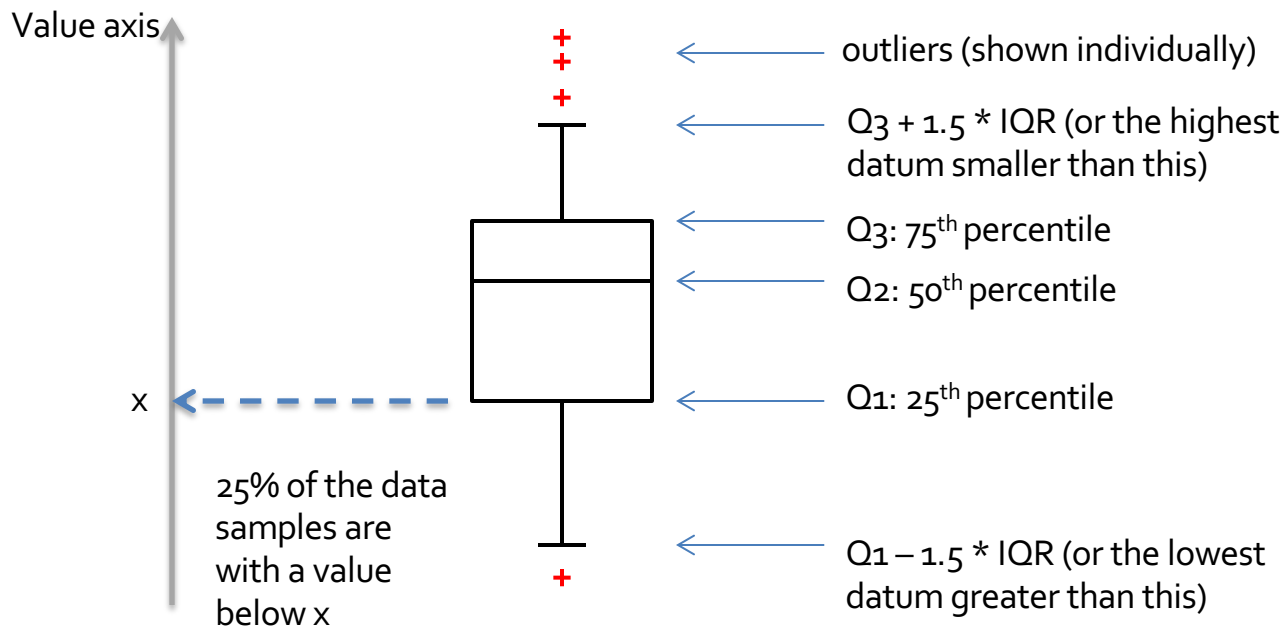
- Conventional histograms / distribution plots might not be space-efficient.



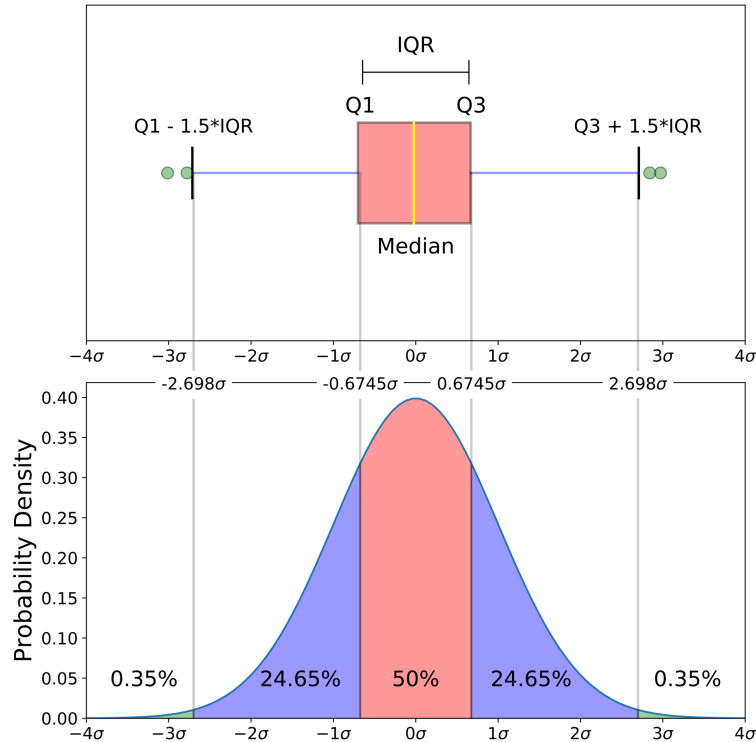
<https://machinelearningmastery.com/data-visualization-in-r/>

Box Plots

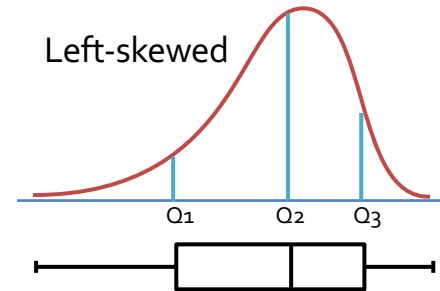
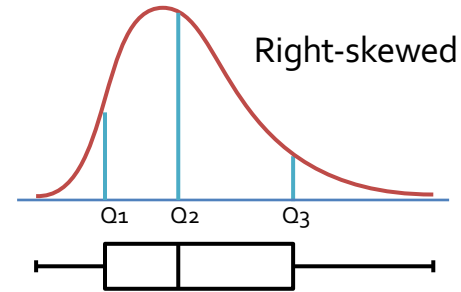
- An efficient way to show **quantitative distribution** of 1D data



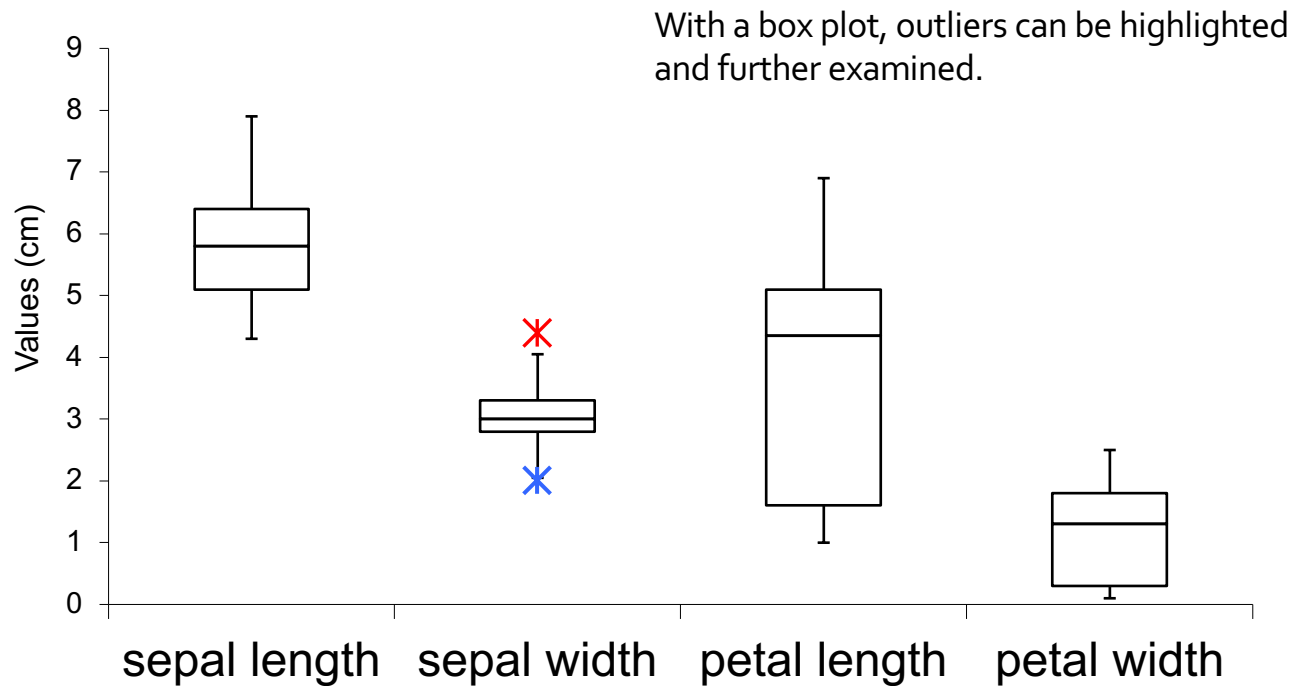
Box Plots



For normal distribution (symmetric)

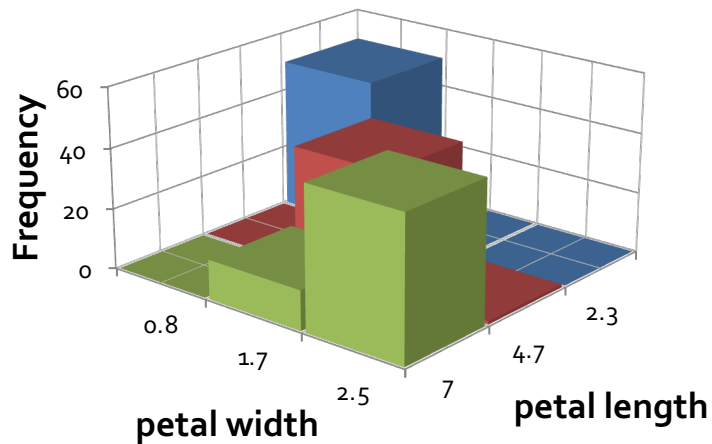


Box Plots



2D Bar Charts

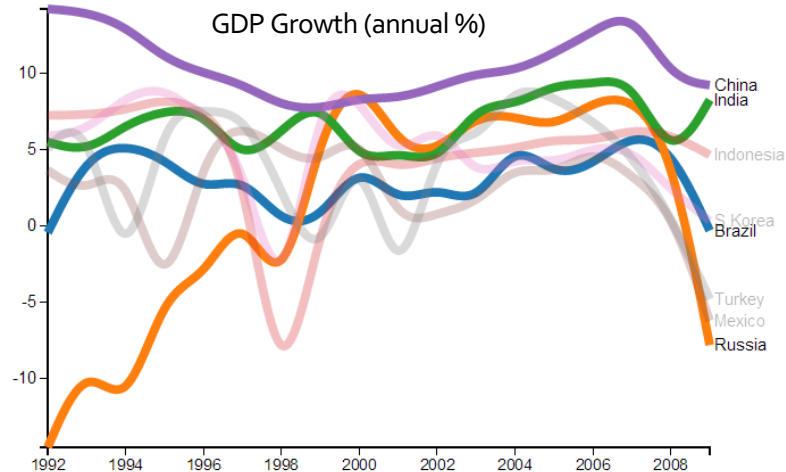
- To show the **joint distribution** of the values of **two variables**



3D effect not good in showing the exact values, but the correlation can be seen clearly

Line Graphs

- Points connected by lines to show how something changes in value (usually over time)

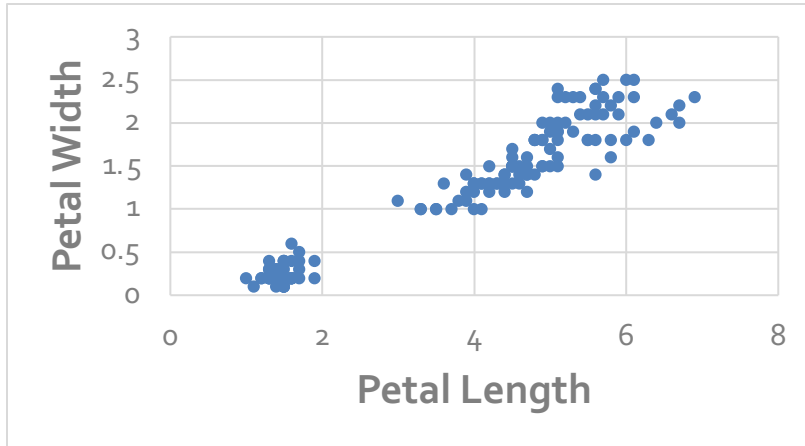


Issues:

- occlusion when too many lines,
- when to use piecewise or smooth lines

Scatter Plots

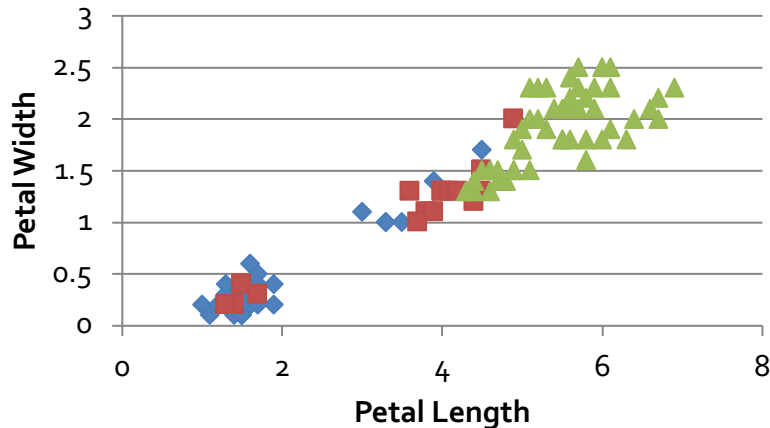
- A plot of points showing the **relationship** between two variables of a set of data



Point position determined by
attribute values

Scatter Plots

- A plot of points showing the **relationship** between two variables of a set of data



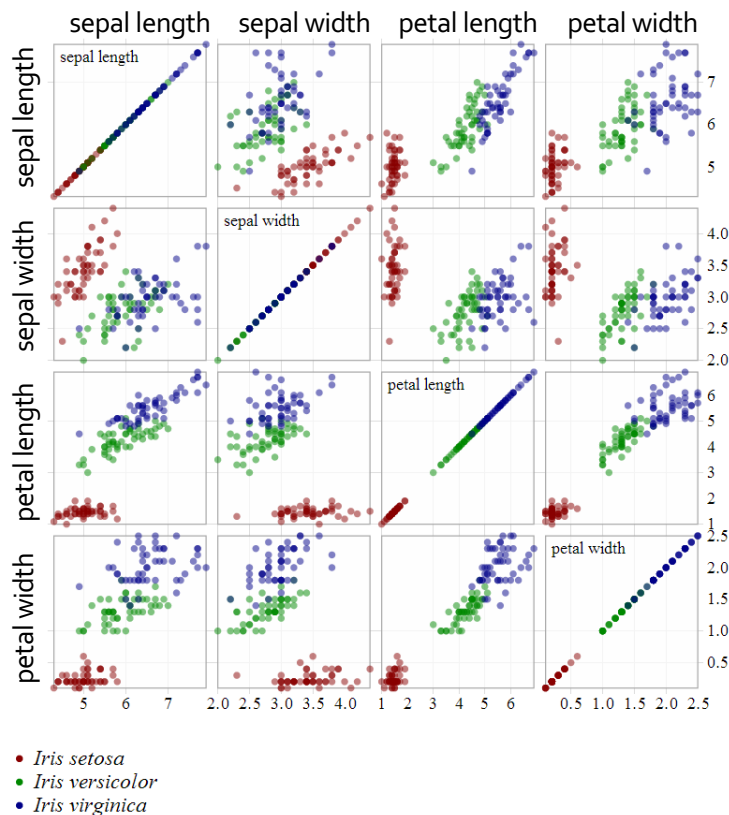
◆ Iris Setosa ■ Iris Versicolour ▲ Iris Virginica

- Point position determined by attribute values
- Additional attributes can be marked by **size**, **shape**, or **color** for each item

Scatter Plots

- 2D scatter plots are commonly used, and there are also 3D scatter plots
- Can compare two attributes at a time only. What if we have a lot of pairwise comparisons to show?
 - have an array (a matrix) of scatter plots

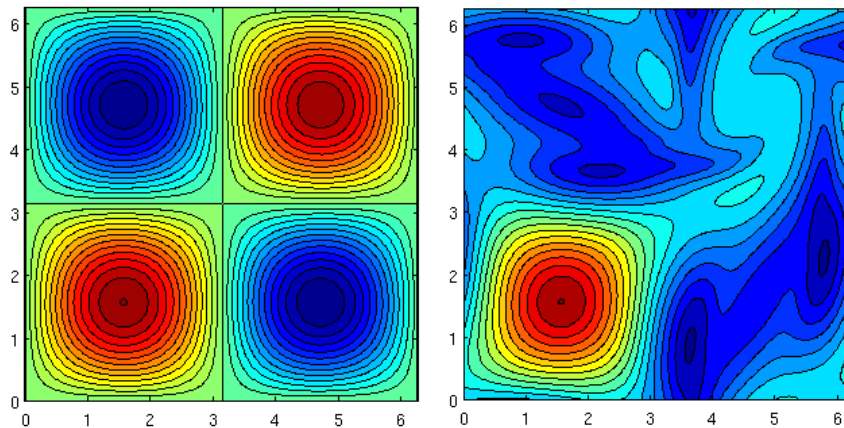
Scatter Plot Matrices



- **Diagonal** plot show the distribution of the 1D data
- Can also make use of the diagonal space for other kinds of 1D visualization, e.g., a histogram
- **Matrix plot** can be used not only for scatter plot, but anything that can deal with a bivariate plot

Contour Plots

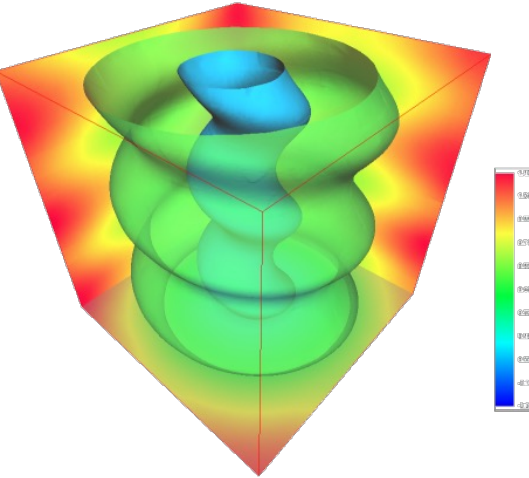
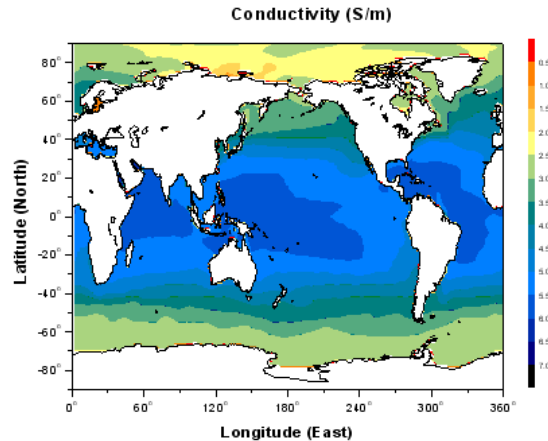
- To show **continuous attributes** measured on a spatial grid
- Partition the space into regions of similar values; boundaries of regions are contour lines called **iso-value** lines, or **isolines**.



[Math, NYU]

Contour Plots

- Commonly used in scientific visualization
- Examples: height fields, temperature, rainfall, etc.



[OriginLab]

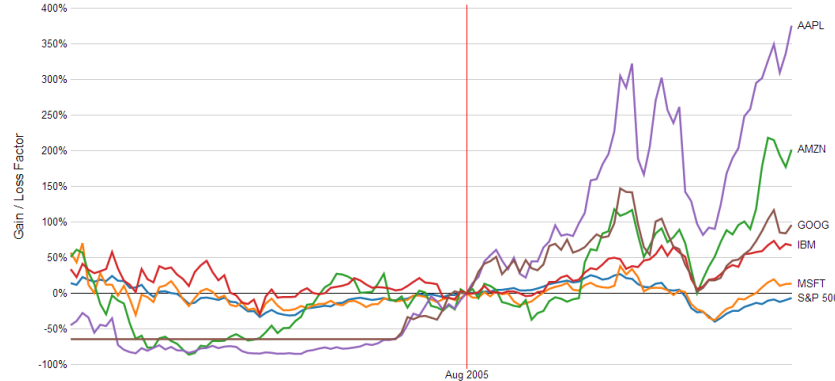
Temporal Data

Time-Series Data

- Set of values that **change over time**
- Examples:
 - Finance (stock prices, exchange rates)
 - Science (temperatures, pollution levels, electric potentials)
 - Public policy (crime rates, public health)
- Common requirements:
 - Able to compare many time series simultaneously
 - Able to use different visualizations in combination

Index Charts

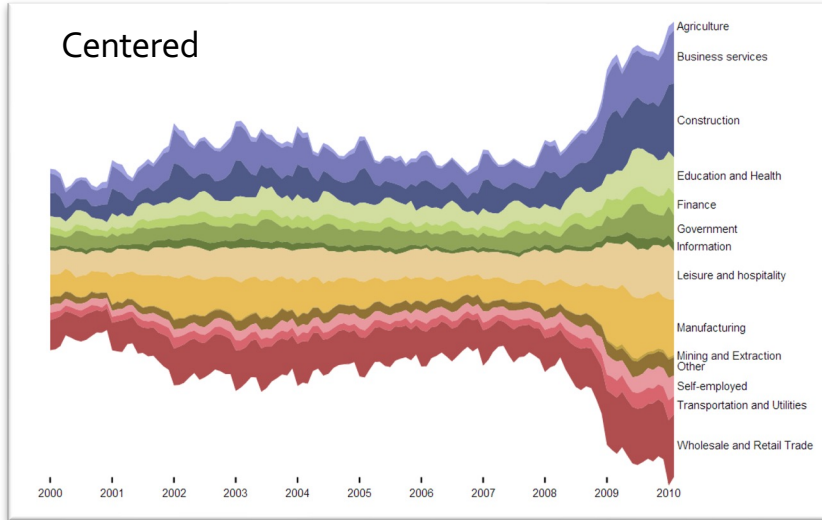
- **Interactive line chart** showing % change based on a selected index point
- Useful for **showing relative changes**



percentage change
of selected stock
prices according to
the day of
purchase

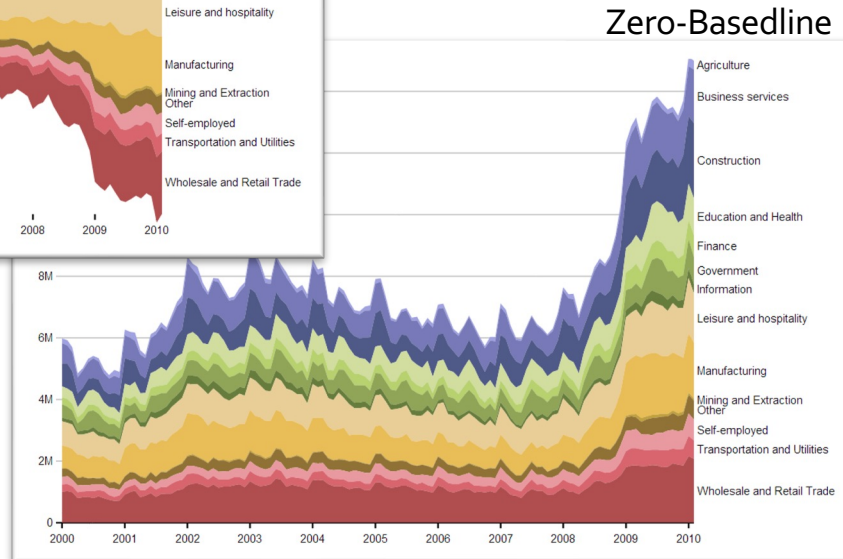
[Heer et al.]

Stacked Graphs



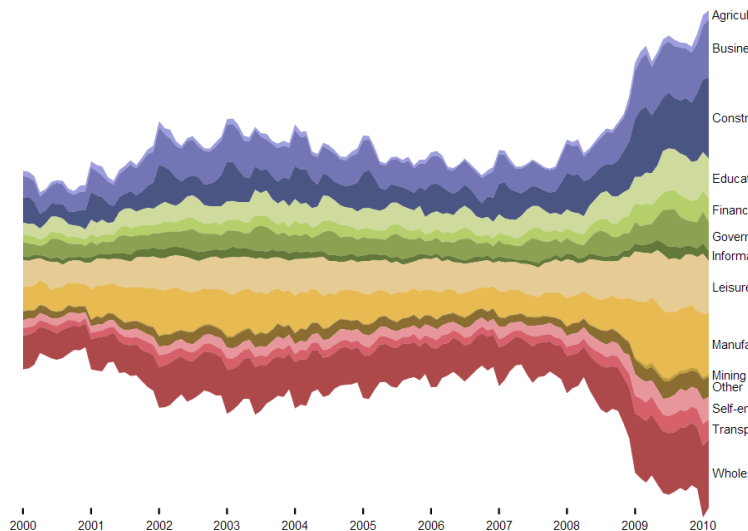
Total counts of unemployed
US workers per industry,
2000-2010.

[Heer et al.]

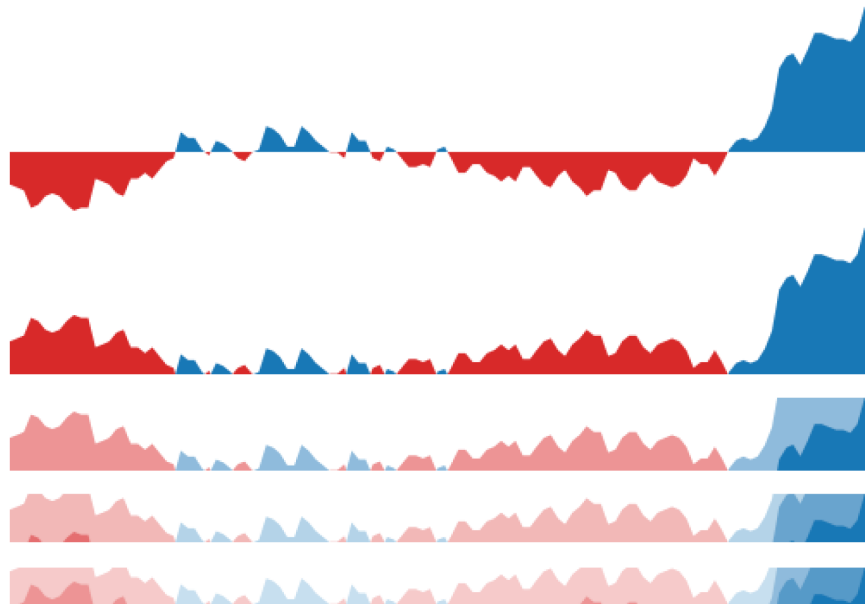


Stacked Graphs

- Stack **area charts** on top of each other
- Useful for showing **summation** of time-series values (aggregation)
- Limitation:
 - negative numbers not supported
 - difficult to interpret trends accurately
 - meaningless for some kind of data (e.g., temperatures)



Horizon Graphs



US unemployment rate, 2000-2010.

Positive values: above average unemployment

Negative values: below average unemployment

[Heer et al.]

[<http://homes.cs.washington.edu/~jheer/files/zoo/ex/time/horizon.html>]

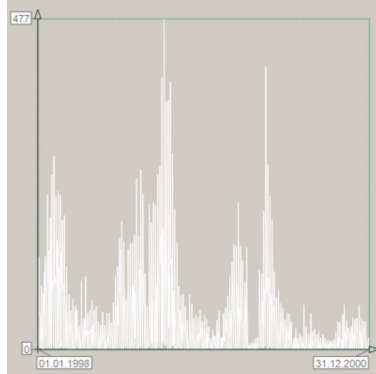
Horizon Graphs

- To divide the area plot into horizontal bands and layer them over each others.
- Useful for increasing the data density (i.e., save space) without sacrificing resolution.
- Limitation: Not intuitive and takes time to learn

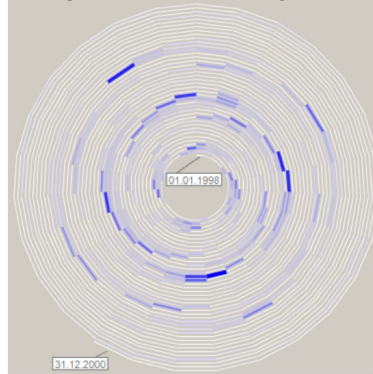
Spiral Graphs

- Use a spirally shaped time axis
- Good for showing or identifying periodic structure of data

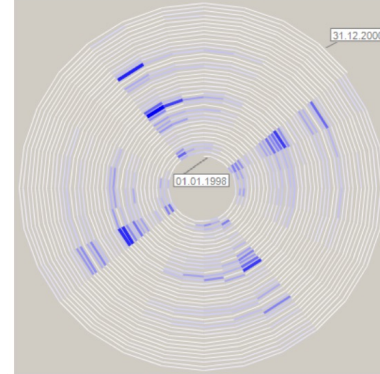
Number of influenza cases over a period of three years



Time series



27 days per cycle



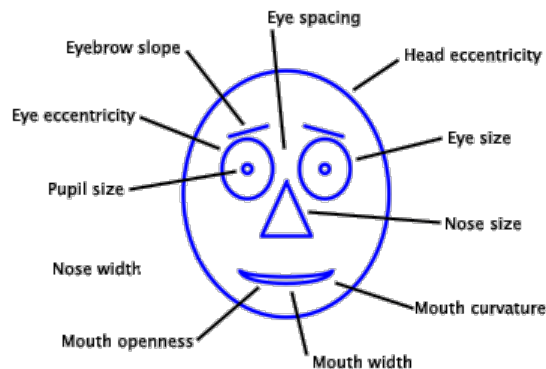
28 days per cycle

[Aigner et al., "Visual Methods for Analyzing Time-Oriented Data", *IEEE TVCG*, 2008.]

Multivariate Data

Chernoff Faces

- Relate data to facial features, something which we find easy to differentiate
- Each feature, e.g., mouth, encode a data dimension by their shape, size, placement and orientation



10 facial features, each corresponds to a parameter in $[0,1]$



All 0



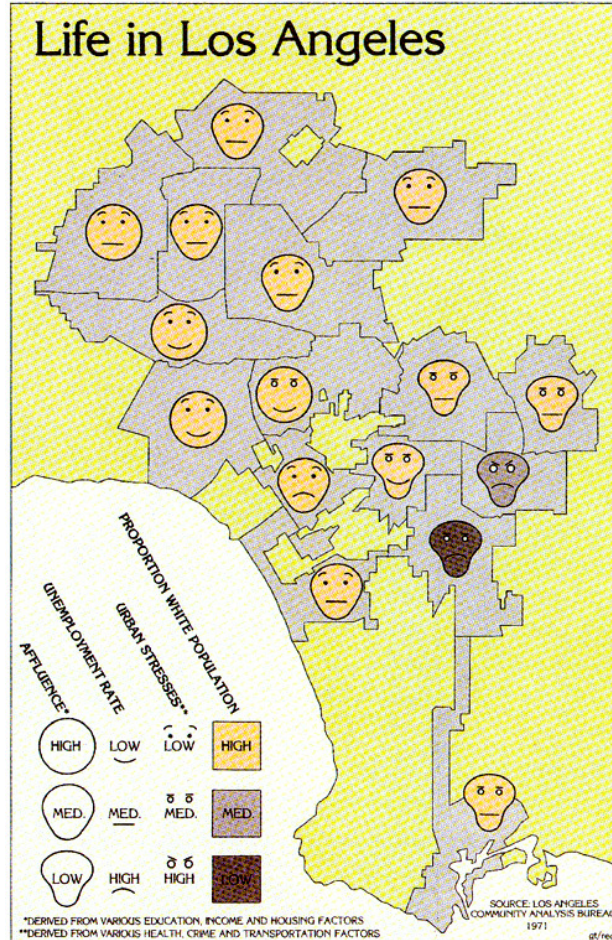
All 0.5



All 1



Random parameters



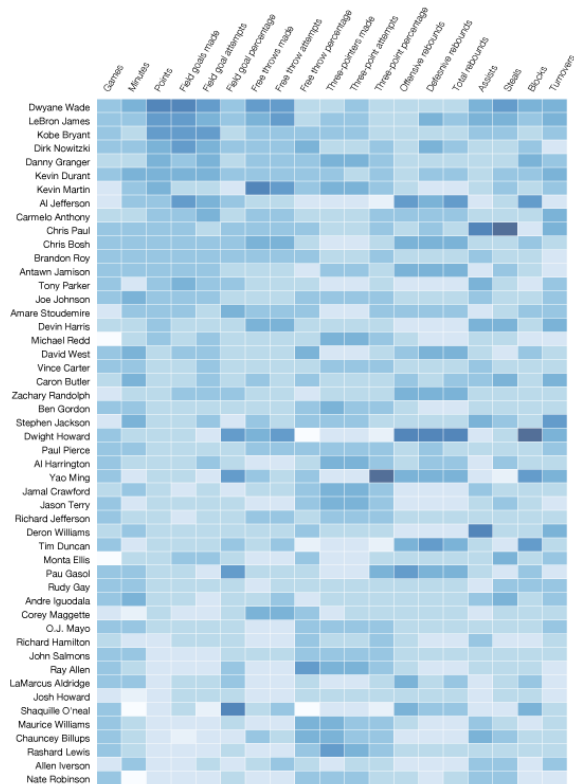
Chernoff Faces

- Represent only **trends** but not actual values
- Drawback: Affected by our perceived importance of a facial feature

Heat Maps

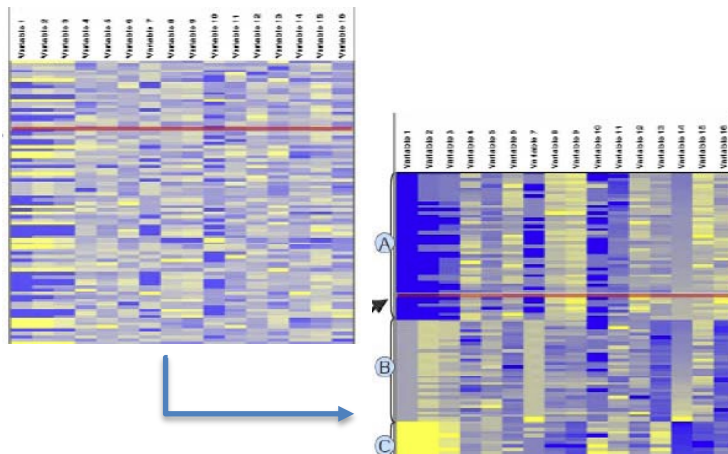
NBA per game performance of top 50 scorers

2008-2009 season

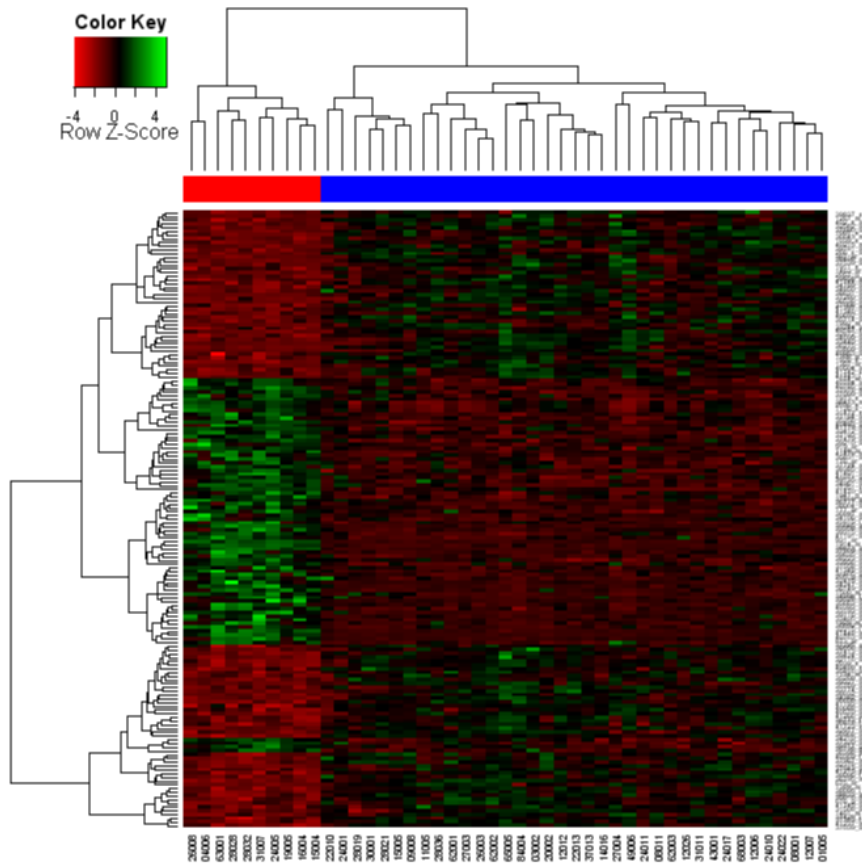


Source: databaseBasketball

- Encode values stored in table entries as colors
- Rows and columns can be reordered to better expose features.



[M. Ward]



column: patient
row: gene

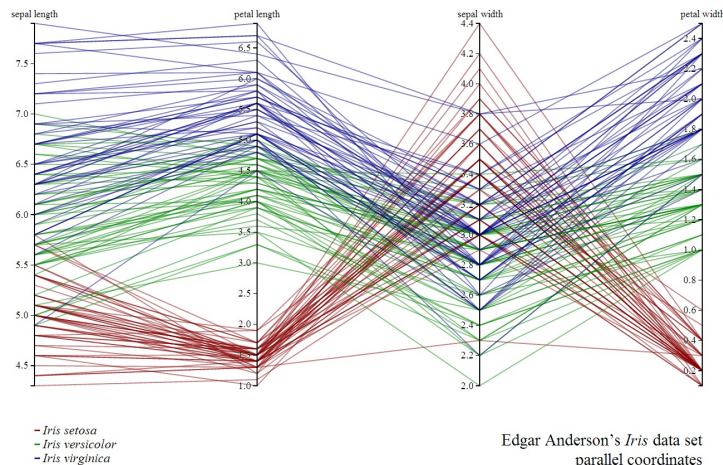
Heatmap from DNA microarray data showing genes expressed differently for two types of leukemia.

[Warwick, http://www2.warwick.ac.uk/fac/sci/moac/people/students/peter_cock/r/heatmap/]

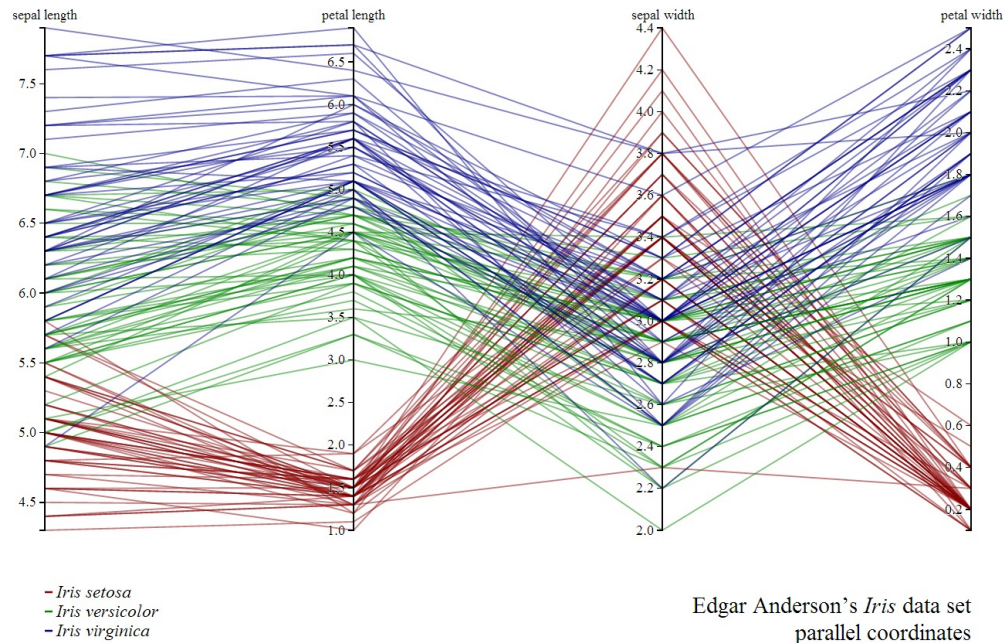
Interactive examples: <http://amp.pharm.mssm.edu/clustergrammer/>

Parallel Coordinates

- How to present all n axes of the n dimensions on a 2D plane?
- Use **parallel axes** instead of orthogonal axes

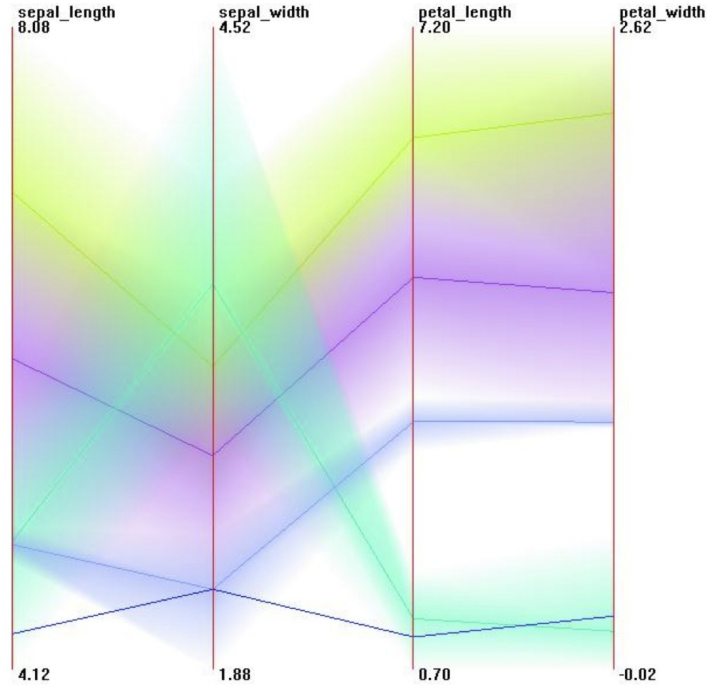


[<http://mbostock.github.io/d3/talk/20111116/iris-parallel.html>]



- Each attribute value of a data item corresponds to a point on a coordinate axis, and the data item is represented as a **polyline** connecting these points
- A distinct class of objects can sometimes be seen as a group of lines on some axes
- Ordering of axes is important for seeing patterns

Parallel Coordinates



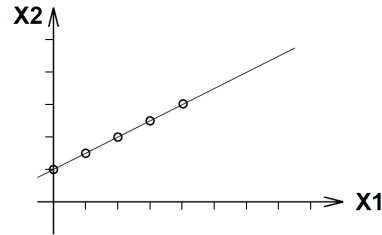
[Ward]

Parallel coordinate plot showing the distribution (i.e., centers and extents) of clusters

Parallel Coordinates

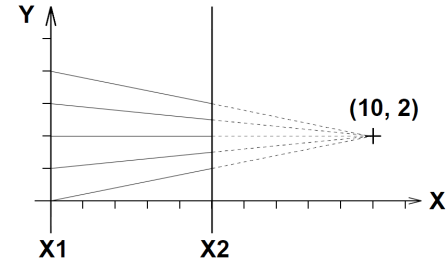
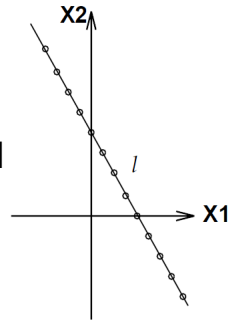
- Parallel correlation

X_1 & X_2
proportional

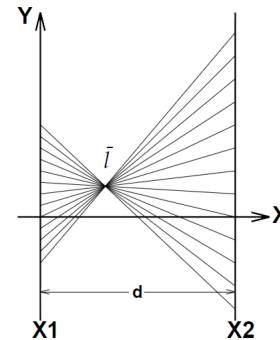


Cartesian point plot

X_1 & X_2
inversely proportional



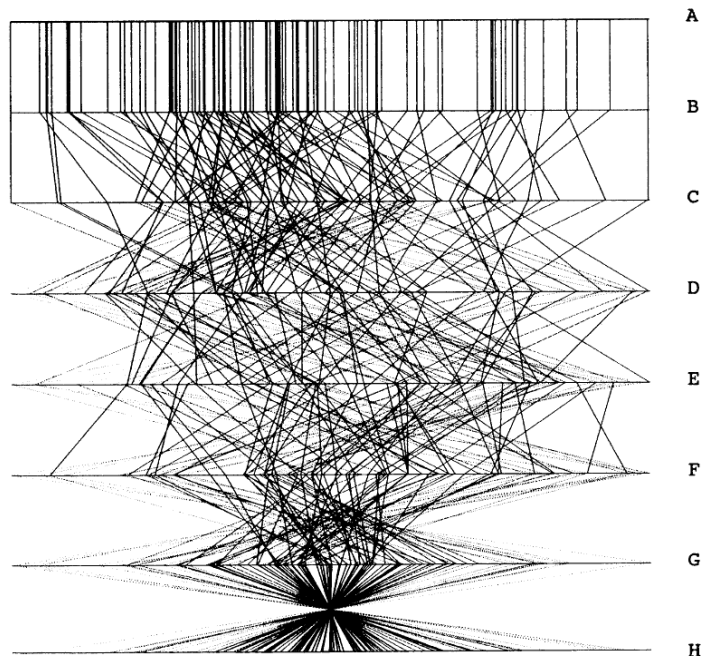
PC plot



[Wong and Bergeron, "30 Years of Multidimensional Multivariate Visualization,"
Scientific Visualization: Overviews, Methodologies & Techniques, 1997.]

Parallel Coordinates

- Parallel correlation

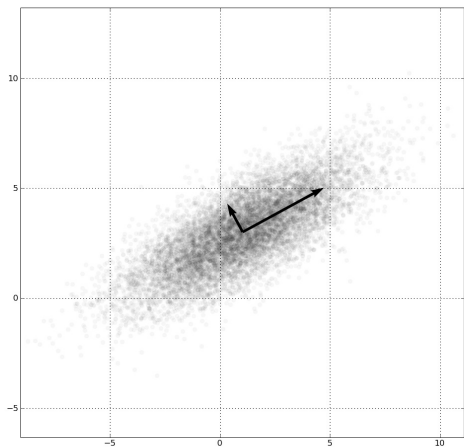


Parallel coordinate plot of six-dimensional data illustrating correlations of ρ (correlation coefficient) = 1, .8, .2, 0, -.2, -.8 and -1.

[Wegman, "Hyperdimensional Data Analysis Using Parallel Coordinates", Journal of the American Statistical Association, 1990.]

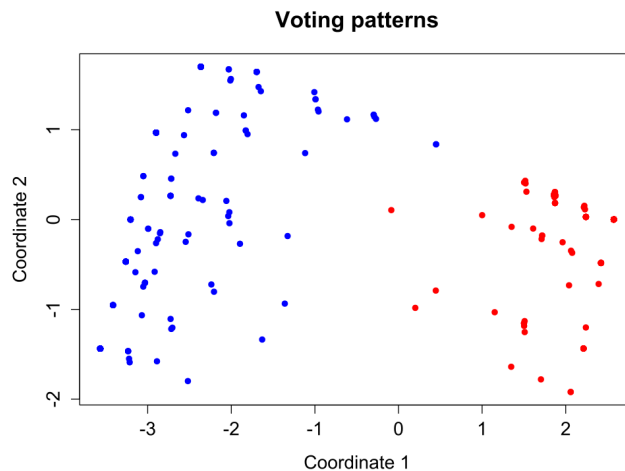
Dimension Reduction

- To remove some of the dimensions out from the display to avoid cluttering
 - Examples: Principle Component Analysis (PCA), Multidimensional Scaling (MDS), Self Organizing Maps (SOM)
- Issue: Resulting dimensions are not the original ones, not intuitive to users



PCA

[<http://commons.wikimedia.org/wiki/File:GaussianScatterPCA.png#mediaviewer/File:GaussianScatterPCA.png>]



MDS

[<http://commons.wikimedia.org/wiki/File:RecentVotes.svg#mediaviewer/File:RecentVotes.svg>]

Dimension Ordering

- Crucial for the effectiveness of many visualization techniques
- Relationship among adjacent dimensions are easier to detect than relationship among those positioned far apart, e.g., Parallel Coordinates, Heat Maps
- Use for attribute mapping to highlight important dimensions, e.g., Chernoff face
- An NP-complete problem equivalent to the Travelling Salesman Problem (TSP)
- Use approximation to compute ordering or by manual ordering (interaction needed)

Visualization Gallery

- Take a look at:
 - Tableau Public
(<https://public.tableau.com/s/gallery>)
 - D3.js
(<http://d3js.org/>)
 - Google Charts
(<https://developers.google.com/chart/interactive/docs/gallery>)
- Try visualize the Iris data set with the different techniques taught in this class using the above tools.
- What can/cannot be done by these tools?

Reference

- Jeffrey Heer, Michael Bostock, and Vadim Ogievetsky. 2010. A tour through the visualization zoo. *Commun. ACM* 53, 6 (June 2010), 59-67. (<http://hci.stanford.edu/jheer/files/zoo/>)
- Matthew Ward, Georges Grinstein and Daniel Keim, "*Interactive Data Visualization: Foundations, Techniques, and Applications*", 2010 [Chapters 6 & 7]